

Chapter Three

Conceptual Database Design (Relational Model)

Chapter 3 - Objectives

- **Database Development Methodology (DDLC)**
- **Phases of Database Design**
- **How to use Entity–Relationship (ER) modelling in database design.**
- **Basic concepts associated with ER model.**

Database Development methodology

- **Information System**
 - **Resources that enable collection, management, control, and dissemination of information throughout an organization.**
 - **Database is fundamental component of IS, and its development/usage should be viewed from perspective of the wider requirements of the organization.**

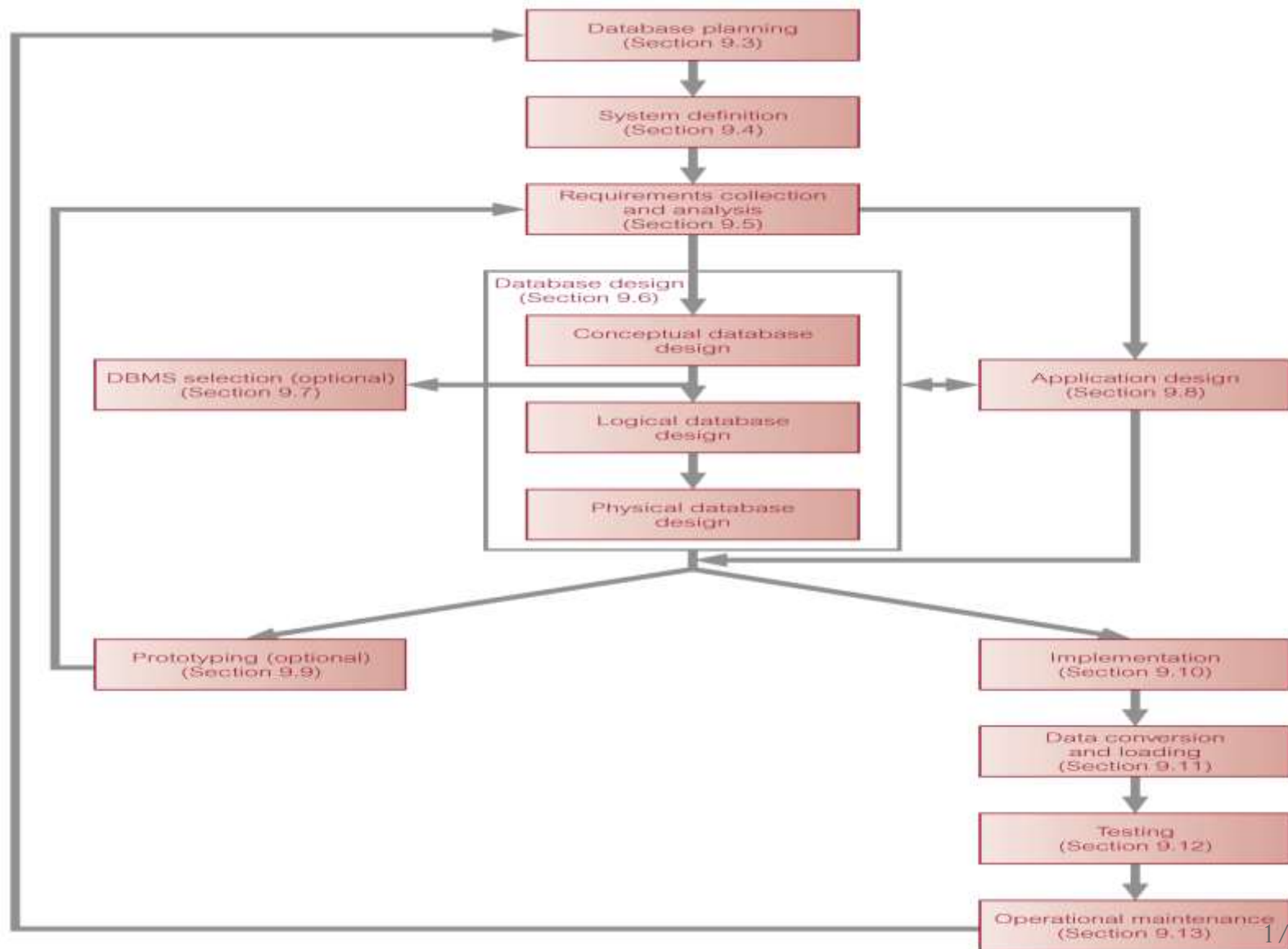
Database System Development Life cycle

- Database planning
- System definition
- Requirements collection and analysis
- Database design
- DBMS selection (optional)

Database System Development Lifecycle Cont'd...

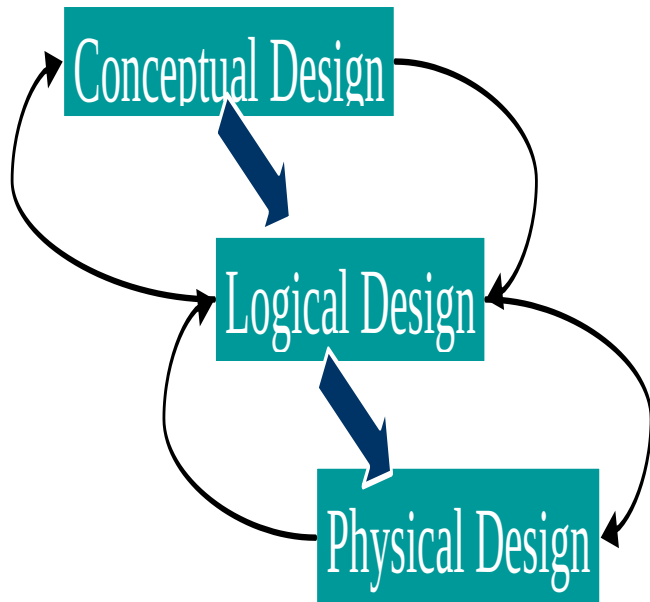
- **Application design**
- **Prototyping (optional)**
- **Implementation**
- **Data conversion and loading**
- **Testing**
- **Operational maintenance**

Stages of the Database System Development Lifecycle Cont'd...



Database Design Methodology

- Three main phases
 - Conceptual database design
 - Logical database design
 - Physical database design



Conceptual Database Design

- The process of constructing a model of the data used in an enterprise, independent of *all* physical considerations.

Logical Database Design

- The process of constructing a model of the data used in an enterprise based on a specific data model (e.g. relational), and dependent of a particular DBMS but other physical considerations.

Physical Database Design

- The process of producing a description of the implementation of the database on secondary storage; it describes the base relations, file organizations, and indexes design used to achieve efficient access to the data, and any associated integrity constraints and security measures.

Conceptual Database Design

- **Conceptual design revolves around discovering and analyzing organizational and user data requirements**

The important activities are to identify

- **Entities**
- **Attributes**
- **Relationships**
- **Constraints**

And based on these components develop the ER model using

- **ER diagrams + Description about the Diagram**

Developing an E-R Diagram

- Designing conceptual model: To identify the entities, attributes, relationships, and constraints on the data,

Different set of methods used during the analysis phase.

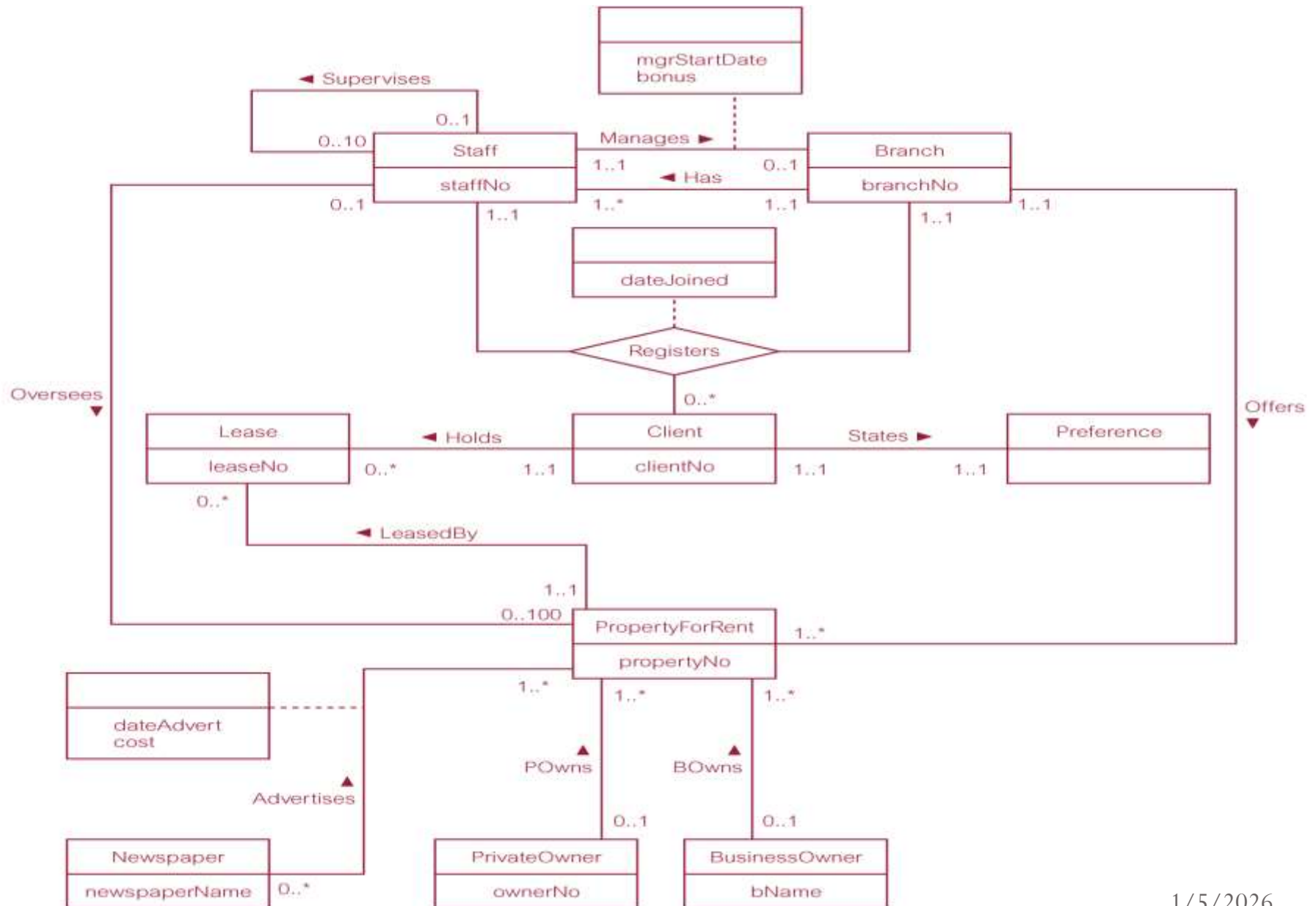
- Information gathering
 - Interviewing end users individually and in a group
 - Questionnaire survey
 - Direct observation
 - Examining different documents like forms, reports, receipts, invoices etc
- Analysis of requirements gathered
 - Nouns -- → prospective entities
 - Adjectives -- → prospective attributes
 - Verbs/verb phrases -- → prospective relationships
 - Commanding Words -- → Constraints

Developing an E-R Diagram Cont'd...

- Develop the basic E-R Model
- Checking for Redundant Relationships in the ER Diagram. Relationships between entities indicate access from one entity to another - this is often referred to as *Navigation* of the ER diagram
- Validate an E-R Model against requirement of the user.
- The basic E-R model presented for review.
- Collecting Feedback from the users

Note: The process is repeated until the end users and designers agree that the E-R diagram is a representation of the organization's activities and functions. (Iterative)

ER diagram of Branch user views of *DreamHome*



Concepts of the ER Model

- **Entity types**
- **Relationship types**
- **Attributes**

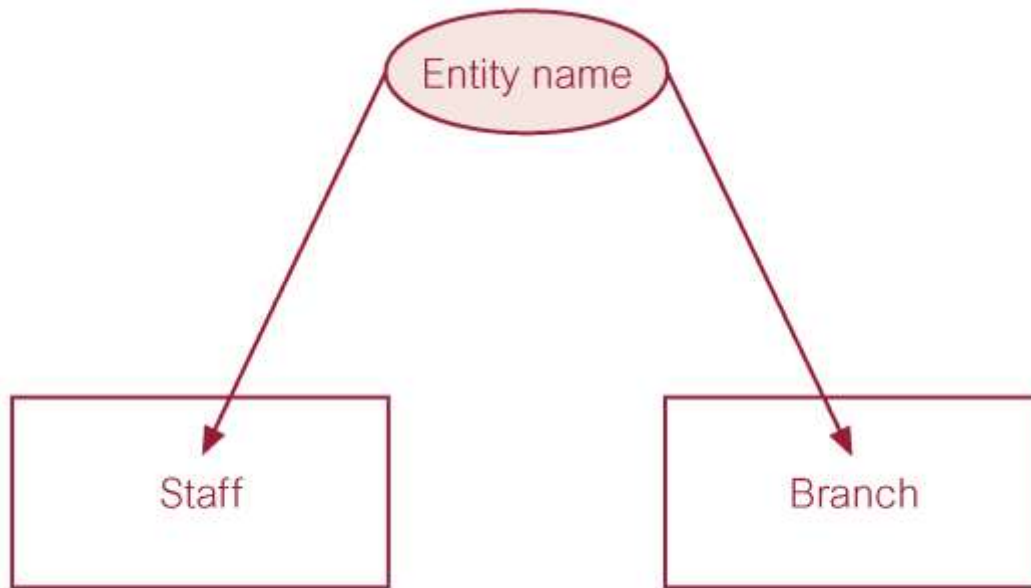
Entity Type

- **Entity type**
 - **Group of objects with same properties, identified by enterprise as having an independent existence.**
- **Entity occurrence**
 - **Uniquely identifiable object of an entity type.**

Examples of Entity Types

Physical existence	
Staff	Part
Property	Supplier
Customer	Product
Conceptual existence	
Viewing	Sale
Inspection	Work experience

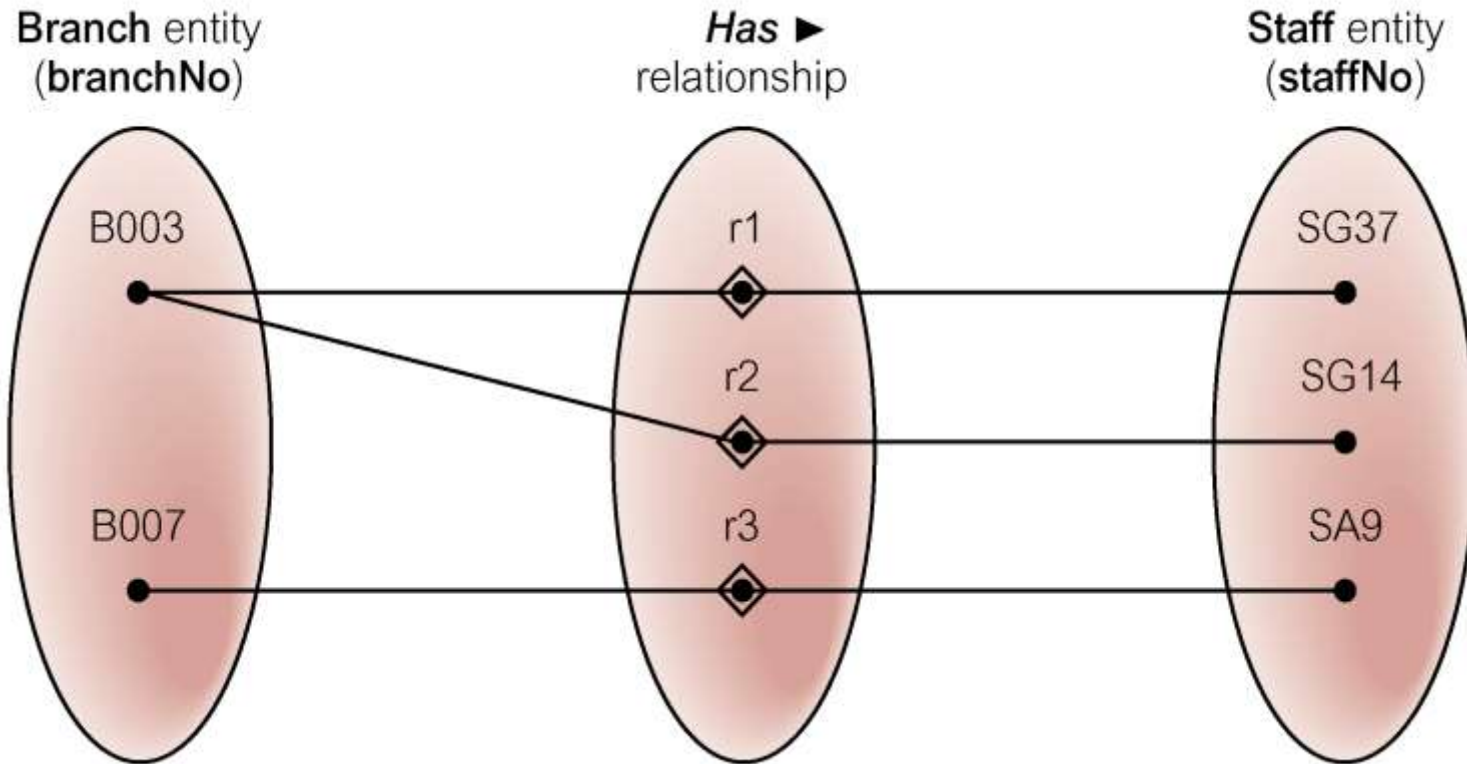
ER diagram of Staff and Branch entity types



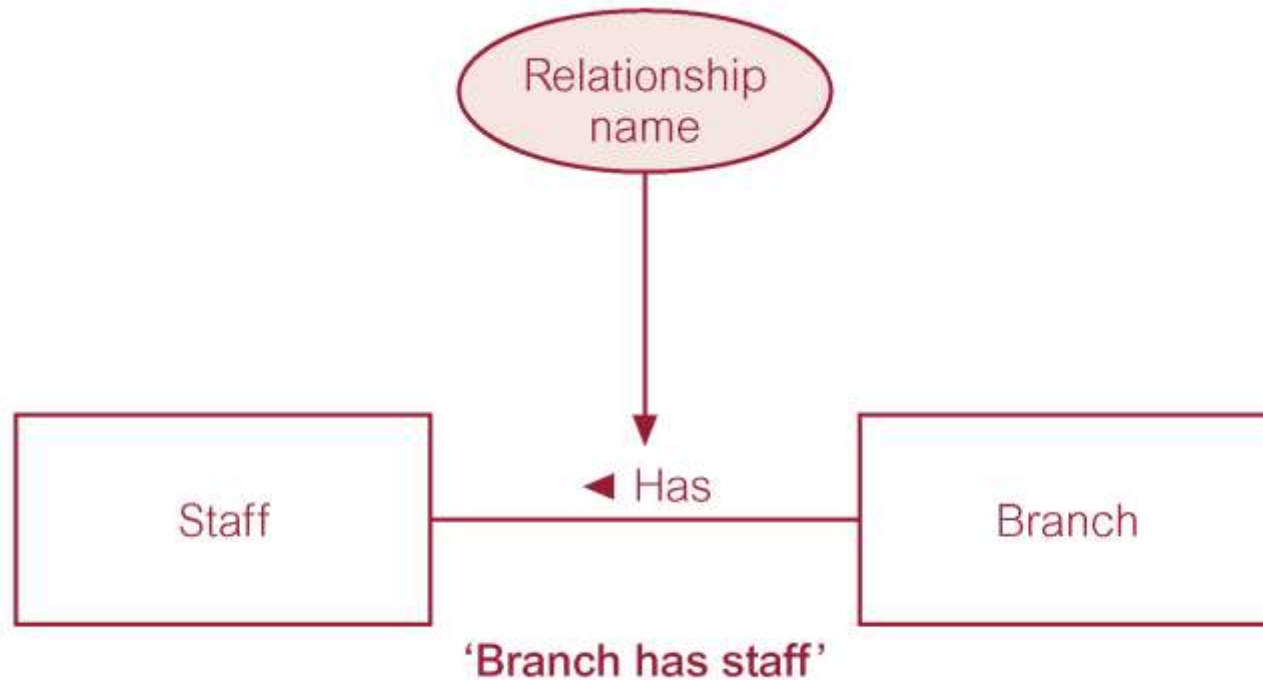
Relationship Types

- **Relationship type**
 - **Set of meaningful associations among entity types.**
- **Relationship occurrence**
 - **Uniquely identifiable association, which includes one occurrence from each participating entity type.**

Semantic net of *Has* relationship type



ER diagram of Branch *Has* Staff relationship



Relationship Types

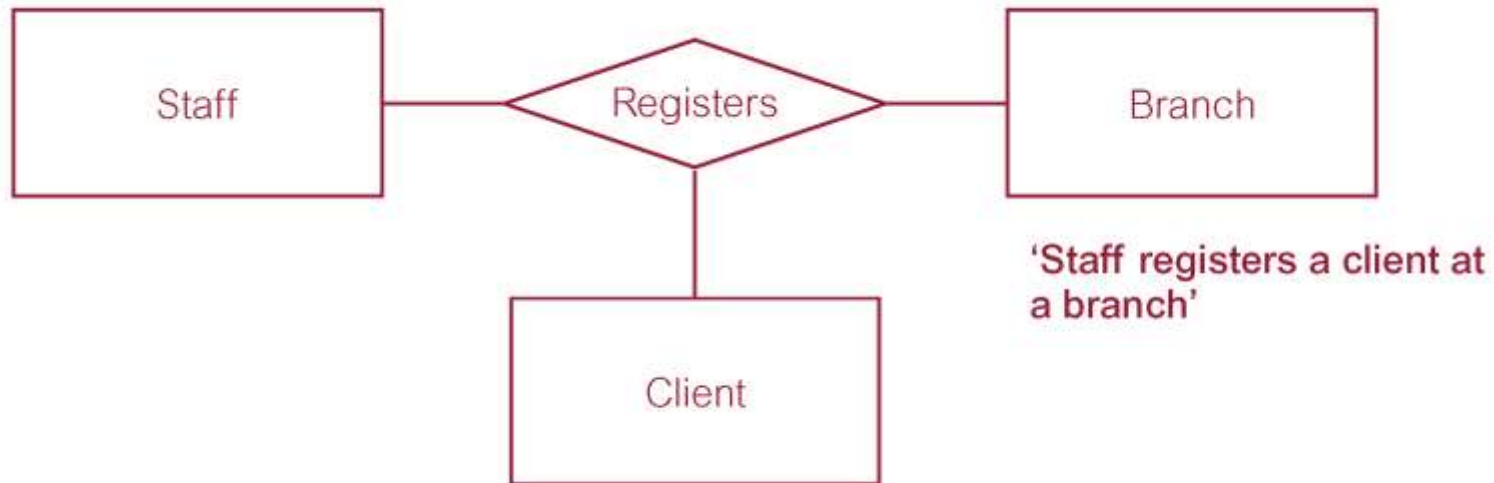
- **Degree of a Relationship**
 - **Number of participating entities in relationship.**
- **Relationship of degree :**
 - **One is Unary**
 - **two is binary**
 - **three is ternary**
 - **four is quaternary**
 - **N is n-ary.**

Binary relationship called *POwns*

'Private owner owns property for rent'



Ternary relationship called *Registers*



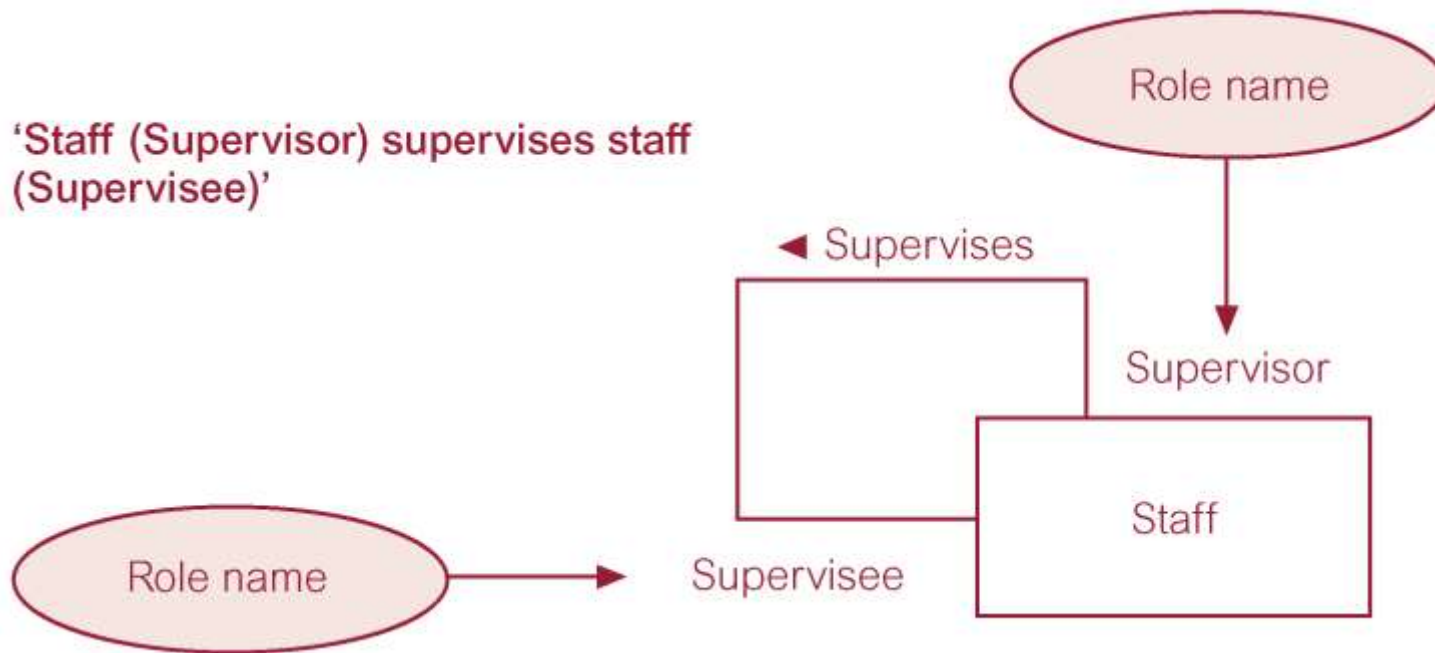
Quaternary relationship called *Arranges*



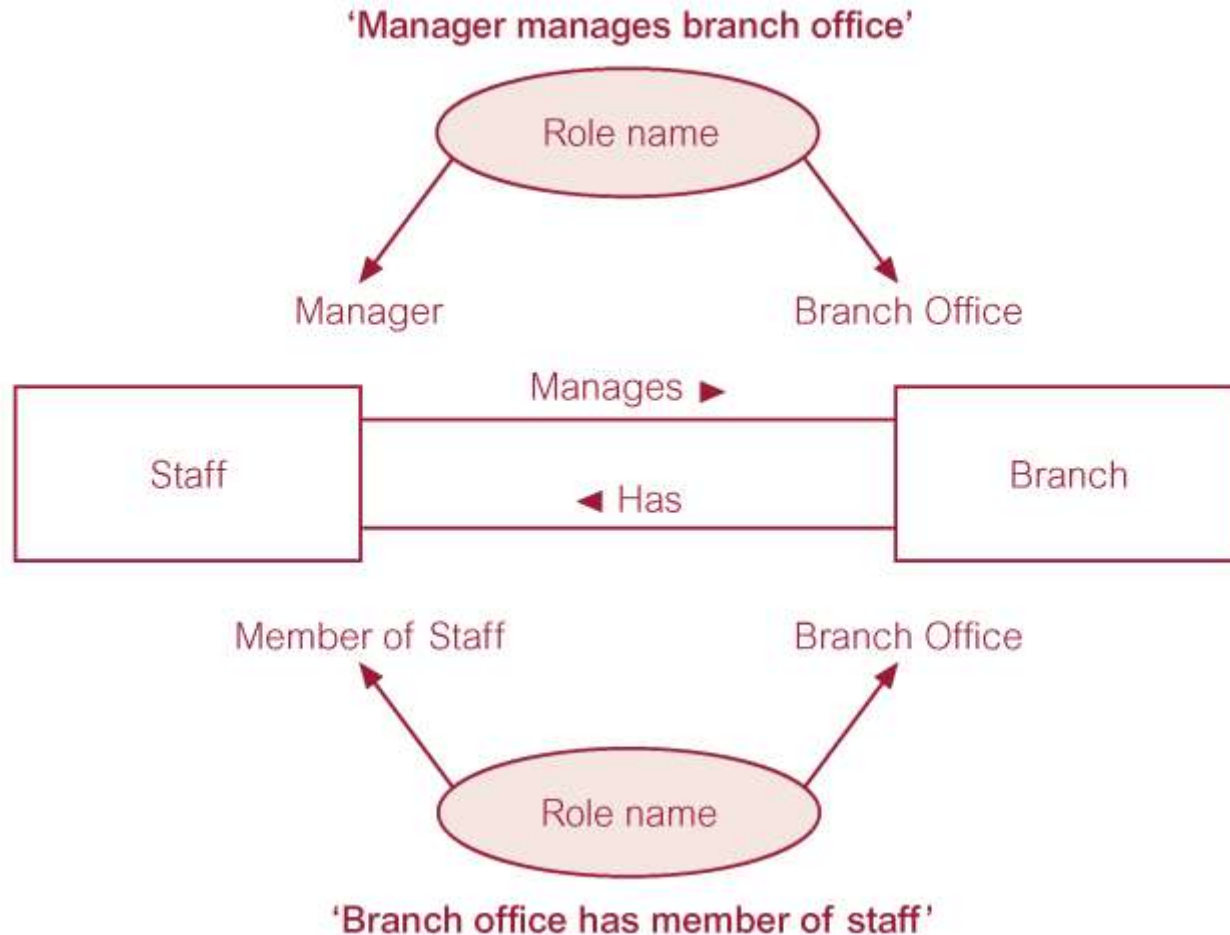
Relationship Types

- **Recursive Relationship**
 - Relationship type where *same* entity type participates more than once in *different roles*.
- Relationships may be given role names to indicate purpose that each participating entity type plays in a relationship.

Recursive relationship called *Supervises* with role names



Entities associated through two distinct relationships with role names



Attributes

- **Attribute**
 - **Property of an entity or a relationship type.**
- **Attribute Domain**
 - **Set of allowable values for one or more attributes.**

Attributes

- **Simple Attribute**
 - **Attribute composed of a single component with an independent existence.**
 - Represented by
 - an Ellipse in traditional E-R model
 - A named field in UML
- **Composite Attribute**
 - **Attribute composed of multiple components, each with an independent existence.**
 - Indicated by indentation in E-R model(UML)
 - Indicated by extension ellipses of an attribute ellipse

Attributes

- **Single-valued Attribute**
 - **Attribute that holds a single value for each occurrence of an entity type.**
- **Multi-valued Attribute**
 - **Attribute that holds multiple values for each occurrence of an entity type.**
 - **Represented by**
 - **double line ellipse in Traditional E-R**
 - **[Min..Max] boundary in UML**

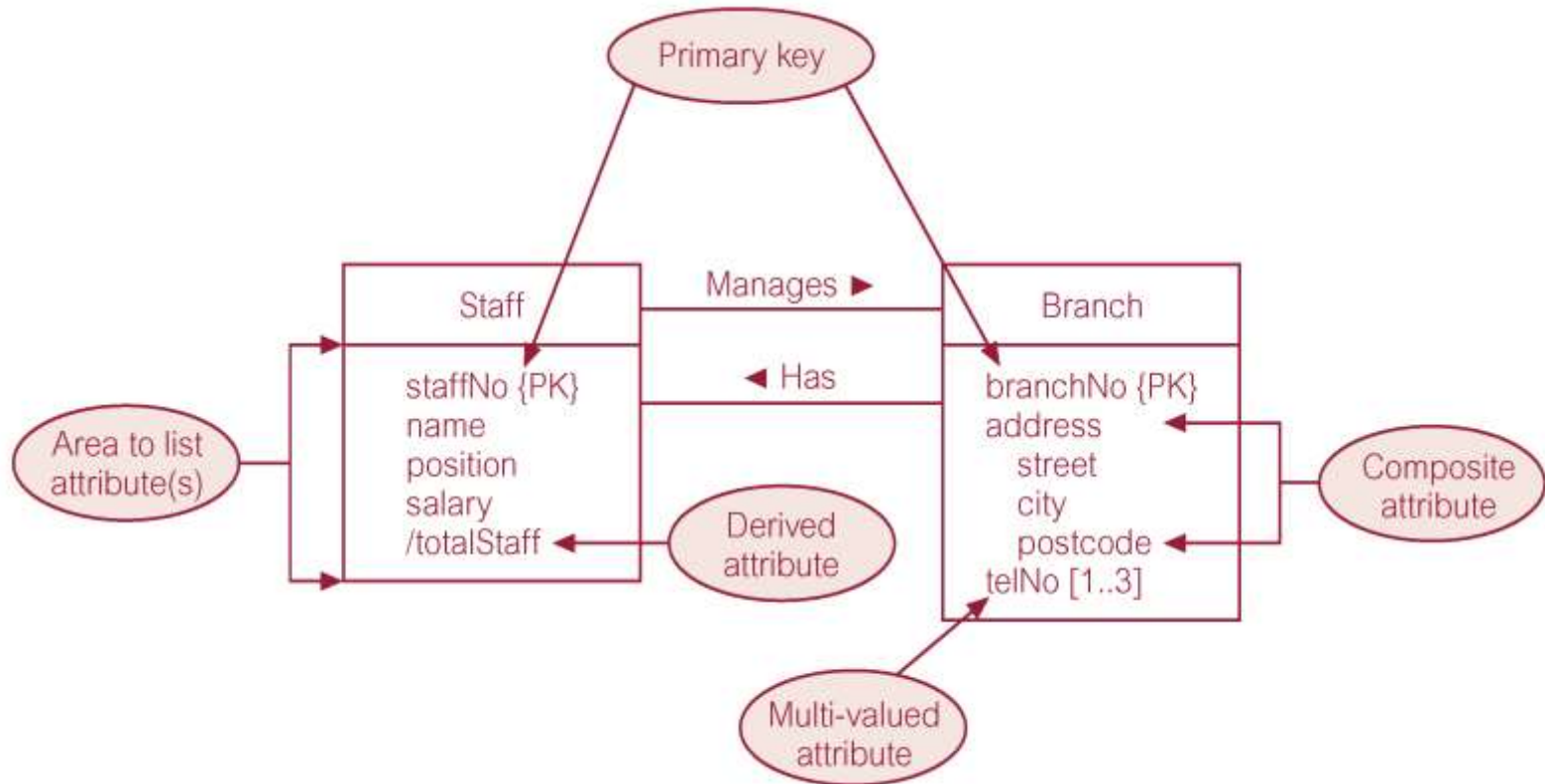
Attributes

- **Derived Attribute**
 - **Attribute that represents a value that is derivable from value of a related attribute, or set of attributes, not necessarily in the same entity type.**
 - **Represented by**
 - a dotted line ellipse in traditional
 - “/” prefix in UML

Keys

- **Candidate Key**
 - **Minimal set of attributes that uniquely identifies each occurrence of an entity type.**
- **Primary Key**
 - **Candidate key selected to uniquely identify each occurrence of an entity type.**
 - Underlined- traditional
 - {Pk} suffix UML
- **Composite Key**
 - **A candidate key that consists of two or more attributes.**
 - Underline all
 - {PK} suffix all

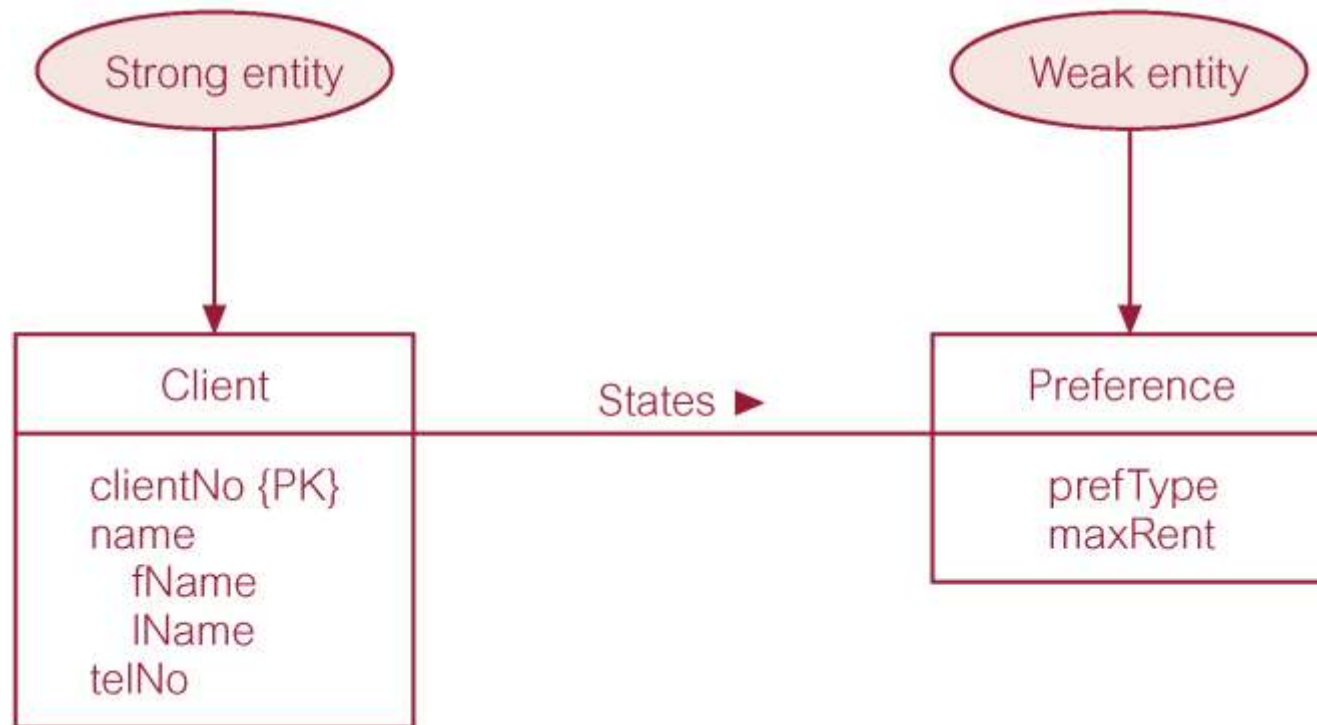
ER diagram of Staff and Branch entities and their attributes



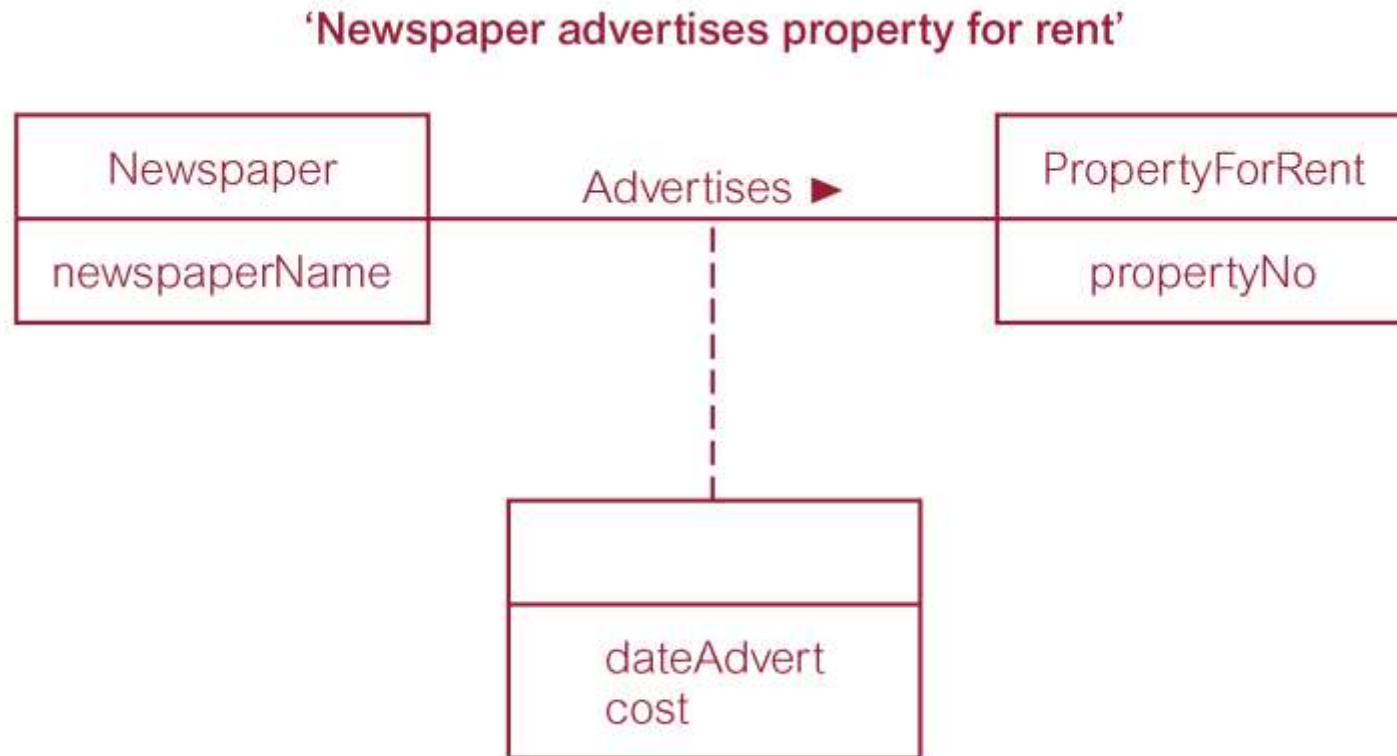
Entity Type

- **Strong Entity Type**
 - Entity type that is *not* existence-dependent on some other entity type.
- **Weak Entity Type**
 - Entity type that is existence-dependent on some other entity type.

Strong entity type called Client and weak entity type called Preference



Relationship called *Advertises* with attributes



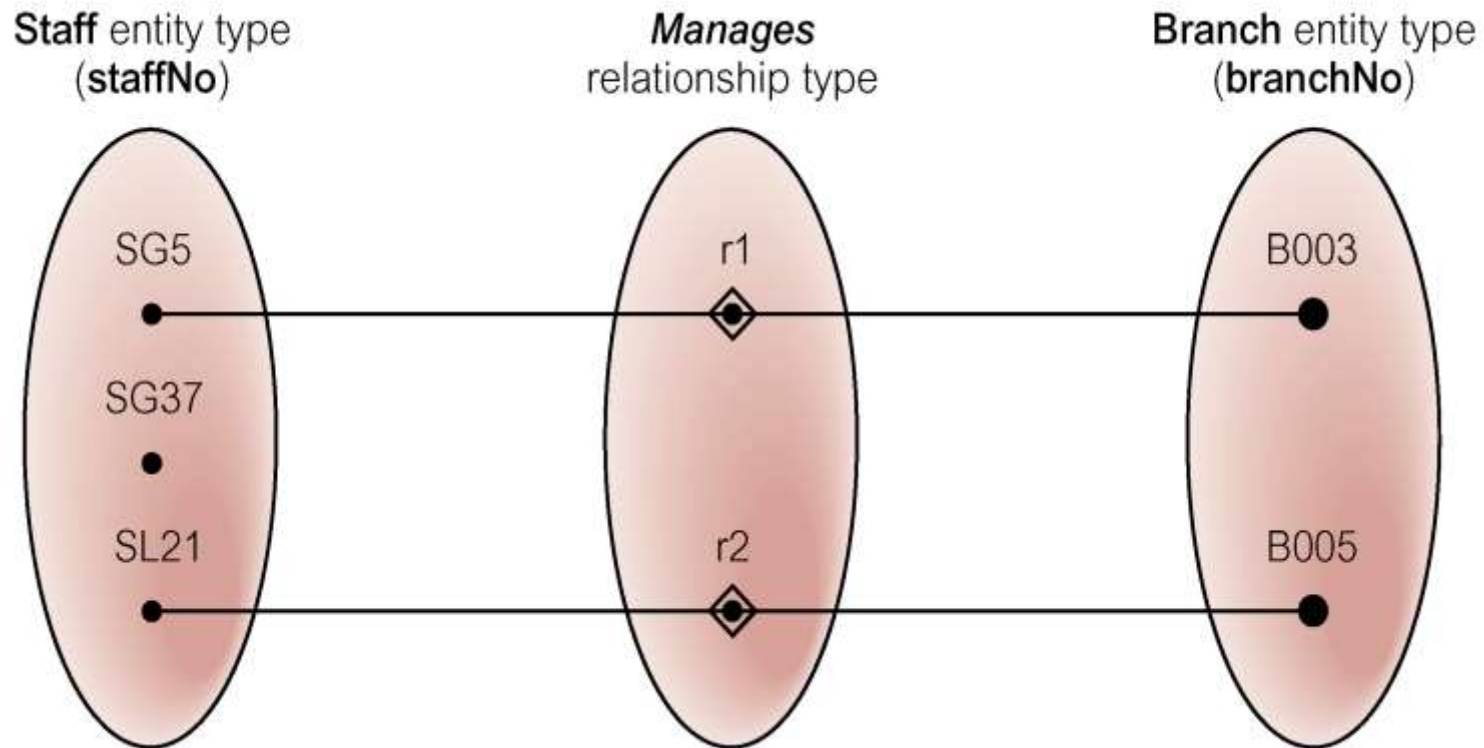
Structural Constraints

- **Main type of constraint on relationships is called *multiplicity*.**
- **Multiplicity - number (or range) of possible occurrences of an entity type that may relate to a single occurrence of an associated entity type through a particular relationship.**
- **Represents policies (called *business rules*) established by user or company.**

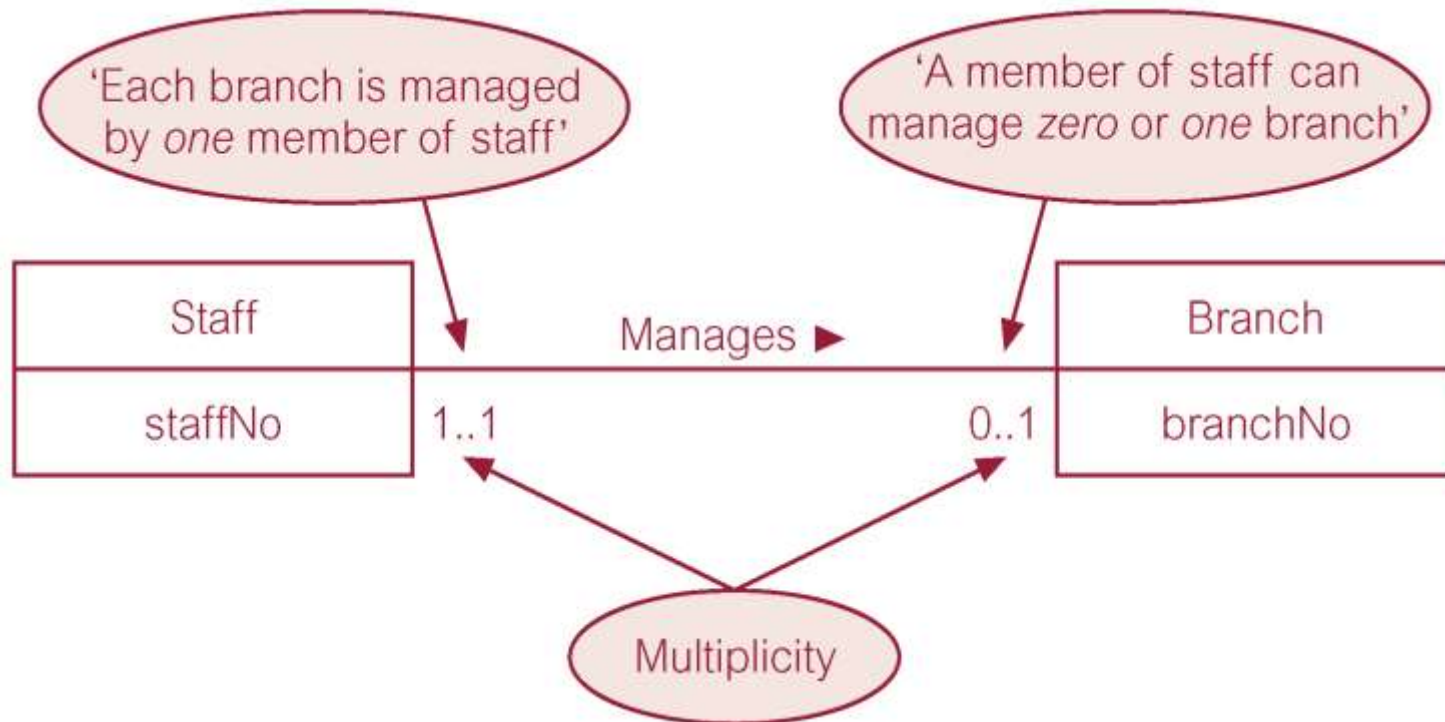
Structural Constraints

- **The most common degree for relationships is binary.**
- **Binary relationships are generally referred to as being:**
 - **one-to-one (1:1)**
 - **one-to-many (1:*)**
 - **many-to-many (*:*)**

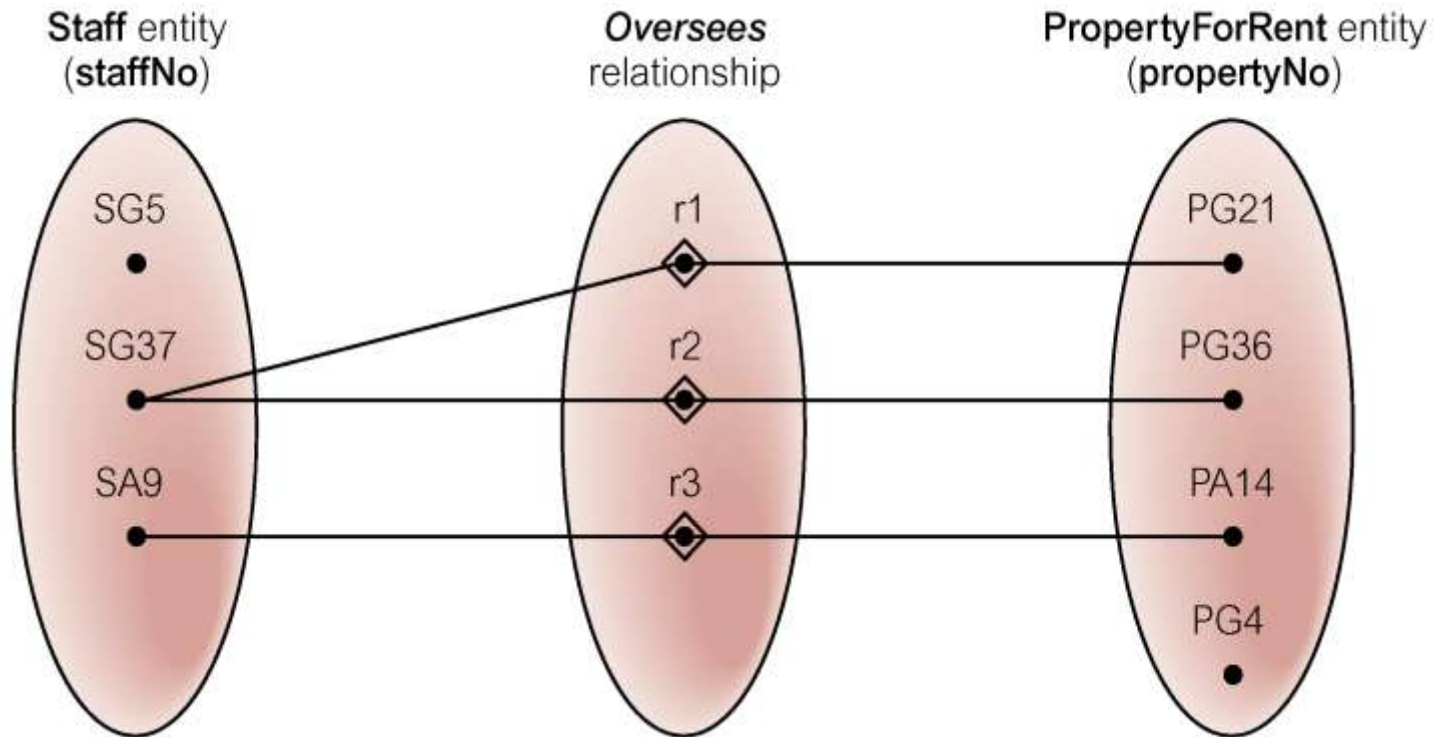
Semantic net of Staff *Manages* Branch relationship type



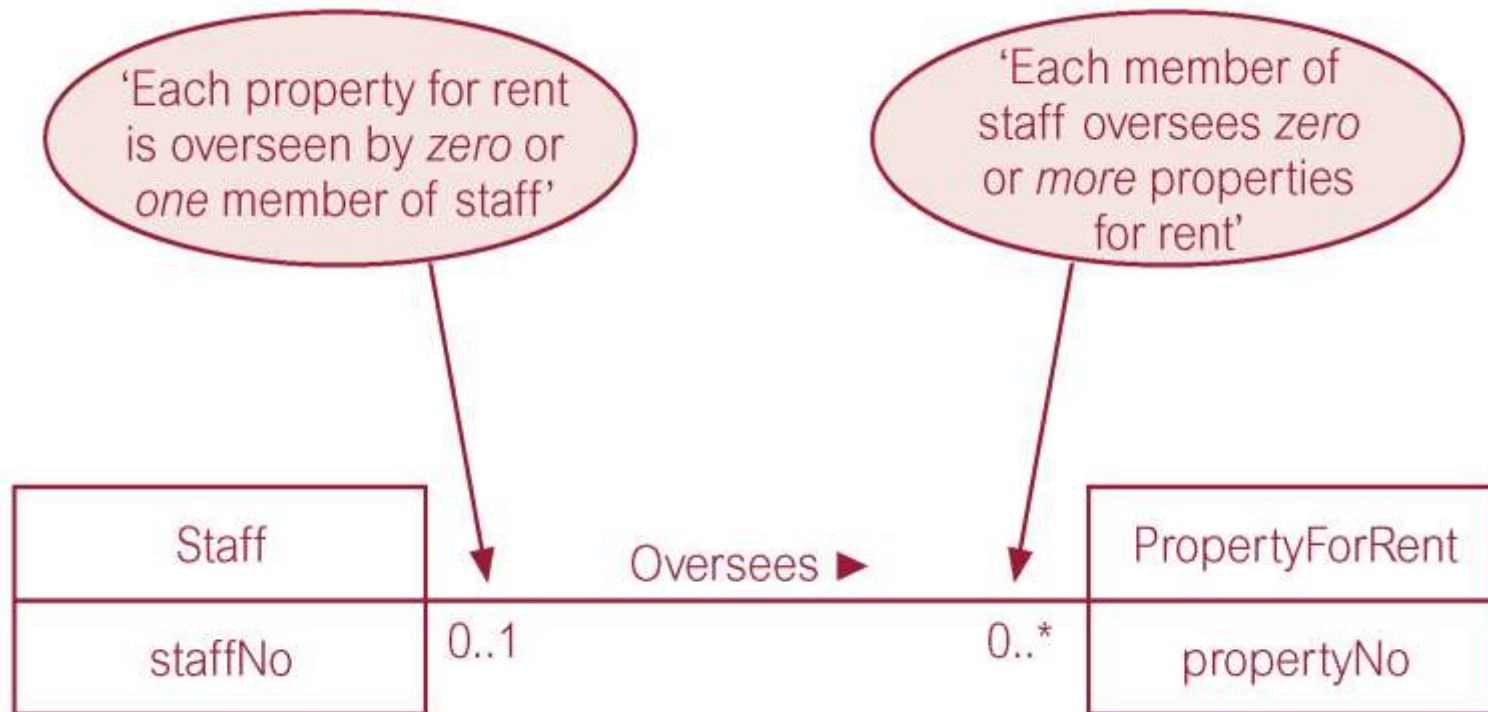
Multiplicity of Staff *Manages* Branch (1:1) relationship



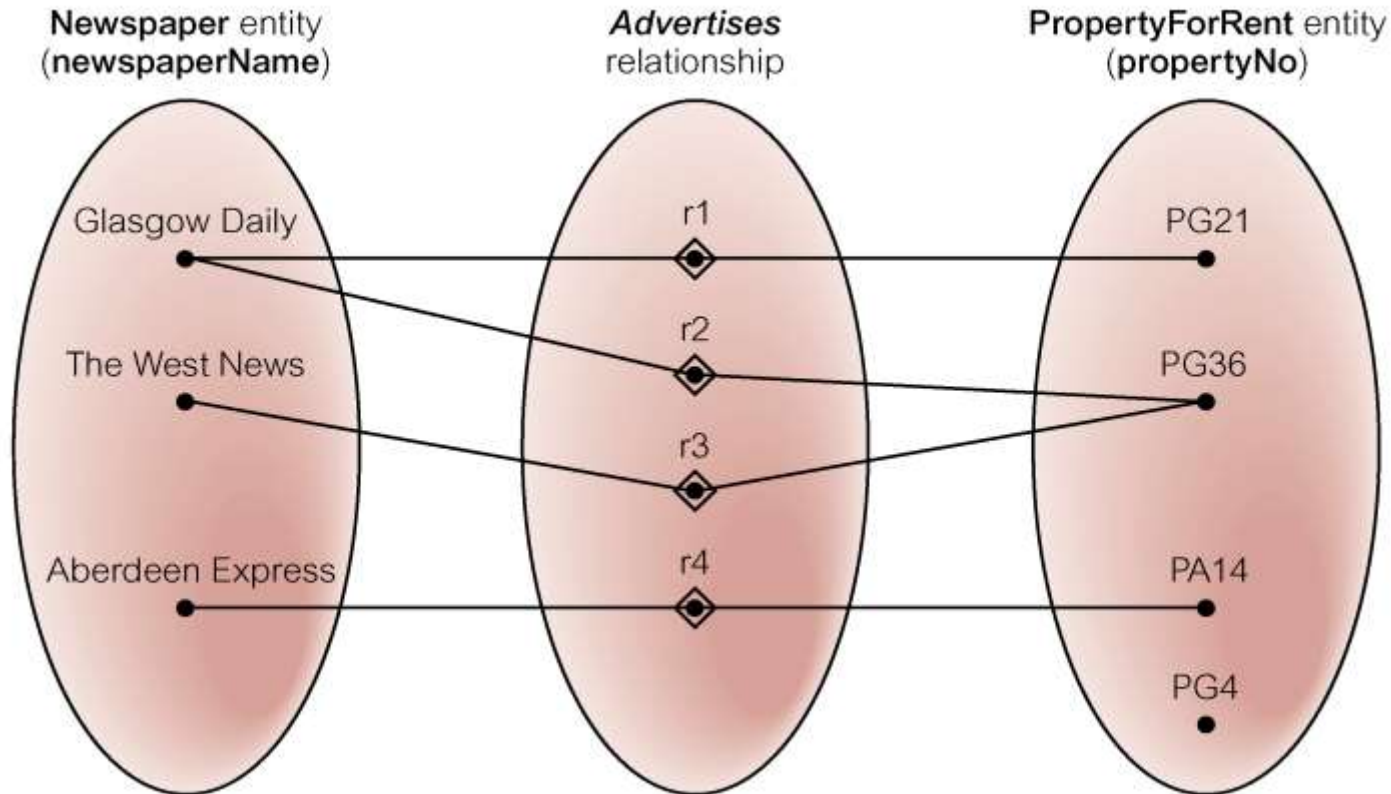
Semantic net of Staff *Oversees* PropertyForRent relationship type



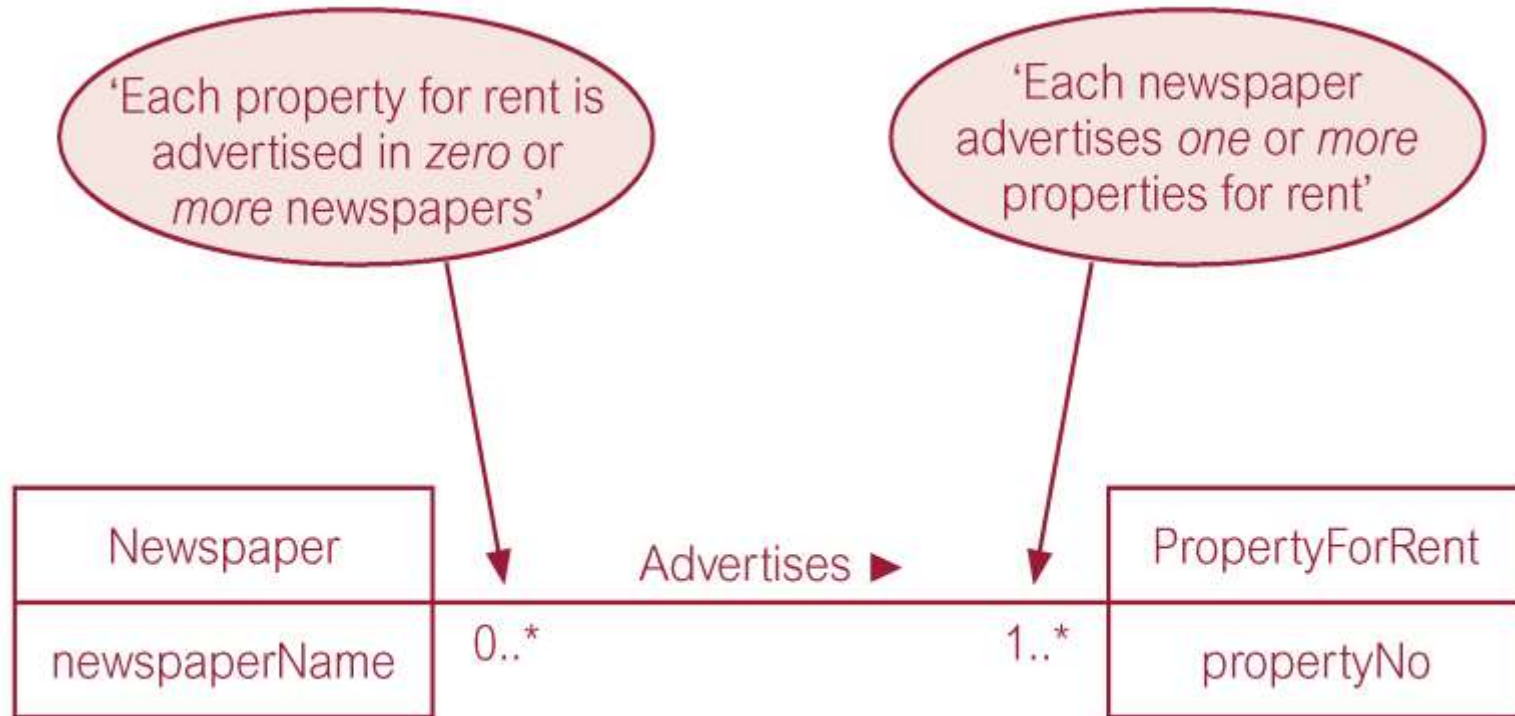
Multiplicity of Staff *Oversees* PropertyForRent (1:*) relationship type



Semantic net of Newspaper *Advertises* PropertyForRent relationship type



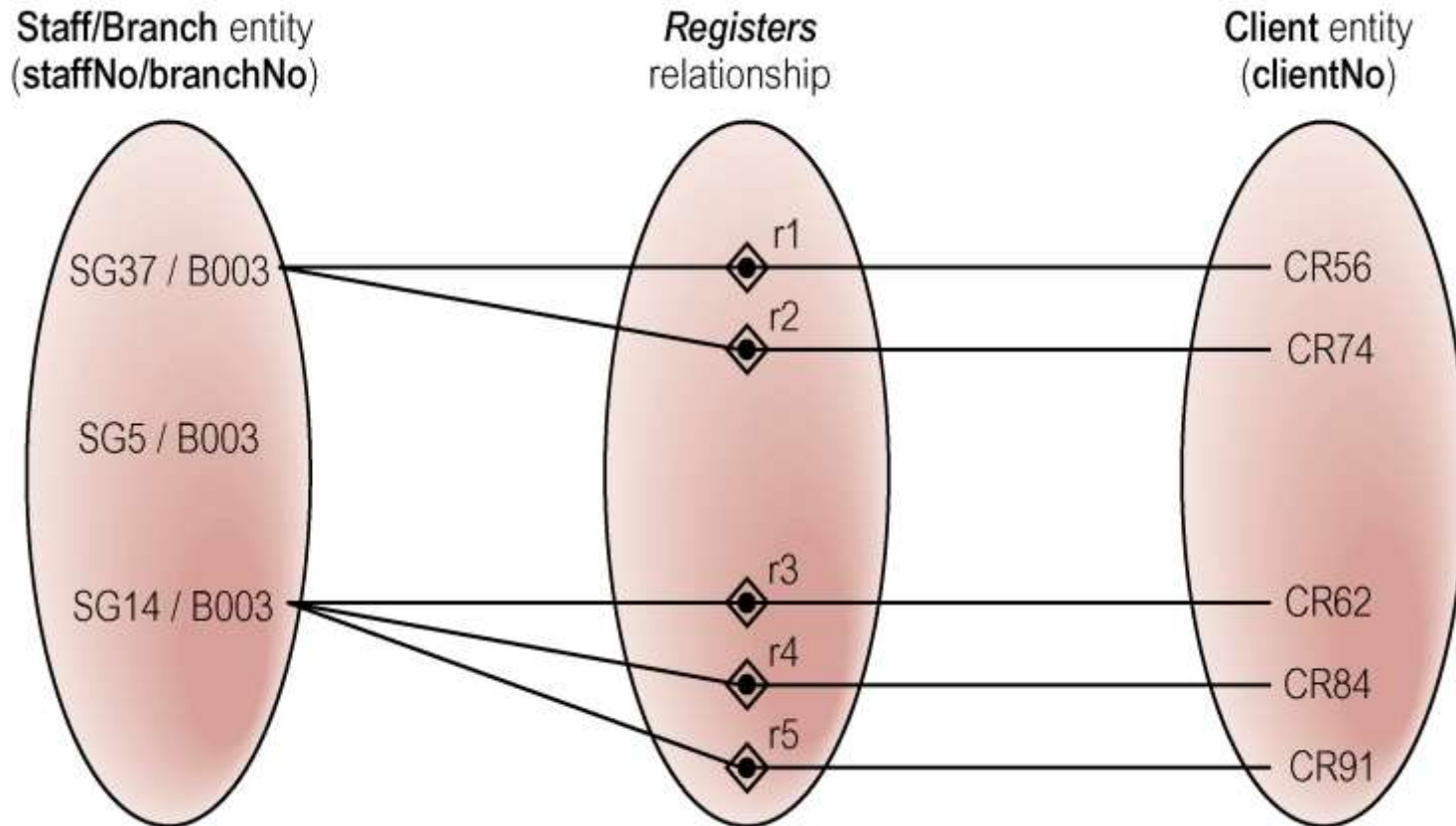
Multiplicity of Newspaper *Advertises* PropertyForRent (*:*) relationship



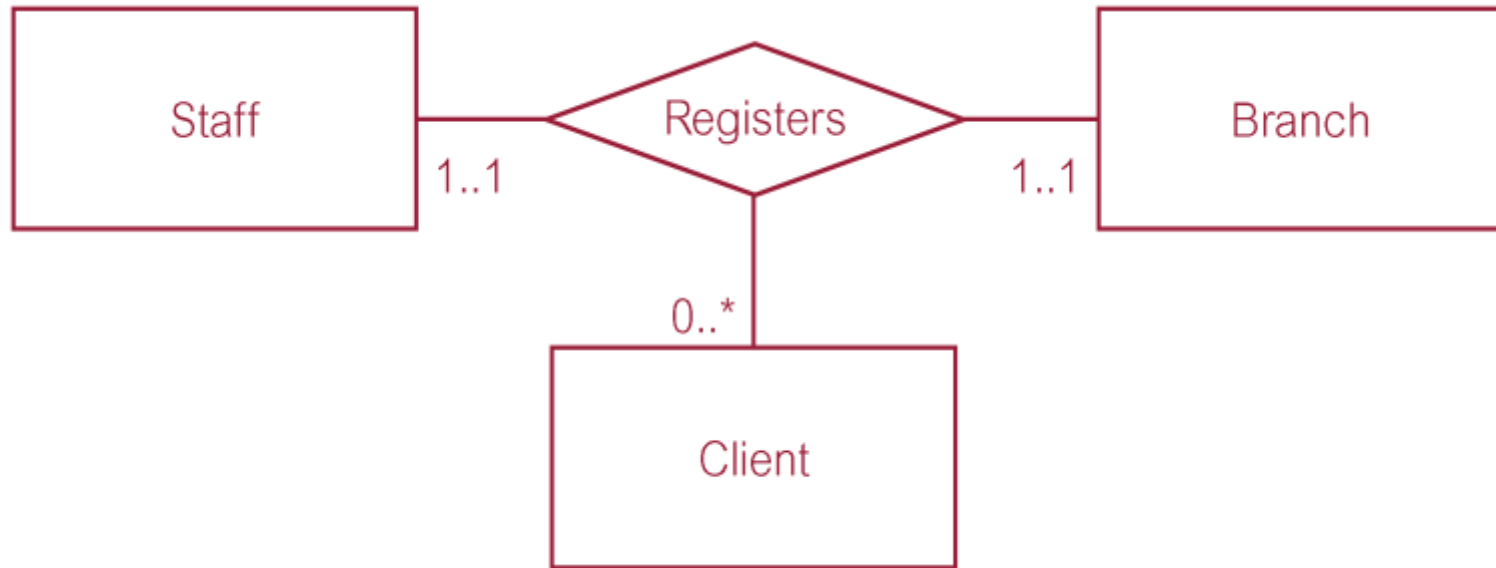
Structural Constraints for Complex Relationships

- **Multiplicity for Complex Relationships**
 - **Number (or range) of possible occurrences of an entity type in an n -ary relationship when other $(n-1)$ values are fixed.**

Semantic net of ternary *Registers* relationship with values for Staff and Branch entities fixed



Multiplicity of ternary *Registers* relationship



Summary of multiplicity constraints

Alternative ways to represent
multiplicity constraints

Meaning

0..1	Zero or one entity occurrence
1..1 (or just 1)	Exactly one entity occurrence
0..* (or just *)	Zero or many entity occurrences
1..*	One or many entity occurrences
5..10	Minimum of 5 up to a maximum of 10 entity occurrences
0, 3, 6–8	Zero or three or six, seven, or eight entity occurrences

Structural Constraints

- **Multiplicity is made up of two types of restrictions on relationships: *cardinality* and *participation*.**

Structural Constraints

- **Cardinality**
 - **Describes maximum number of possible relationship occurrences for an entity participating in a given relationship type.**
- **Participation**
 - **Determines whether all or only some entity occurrences participate in a relationship.**

Multiplicity as cardinality and participation constraints

